DATA-380 Final Project: Cost of Living in the United States

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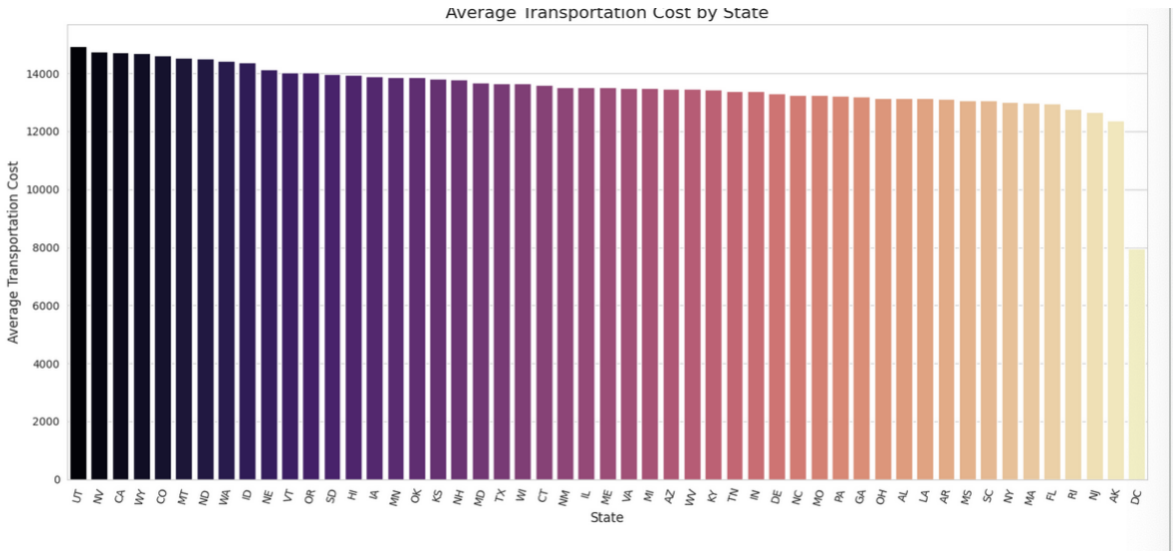
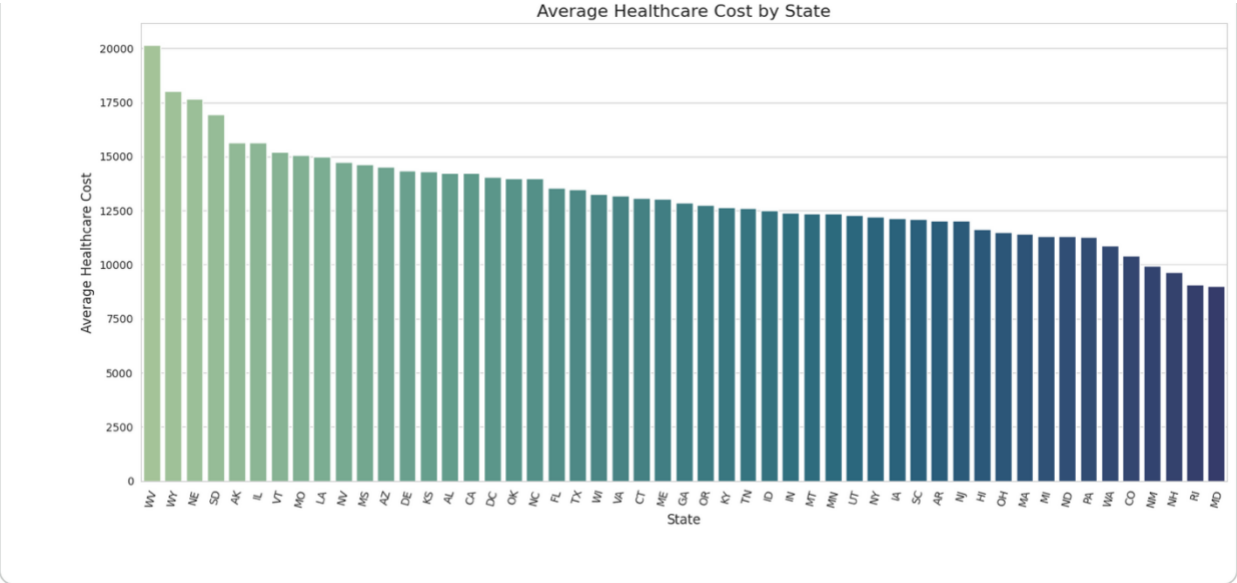
Quantitative Variables	**Categorical Variables**
Num_adults	State
Num_children	Family_member_count
Total_cost	isMetro
Housing_cost	Region
Food_cost	
Transportation_cost	
Healthcare_cost	
Childcare_cost	
Taxes	
Median_family_income	
Total_family_members	

Average Transportation Cost by State:



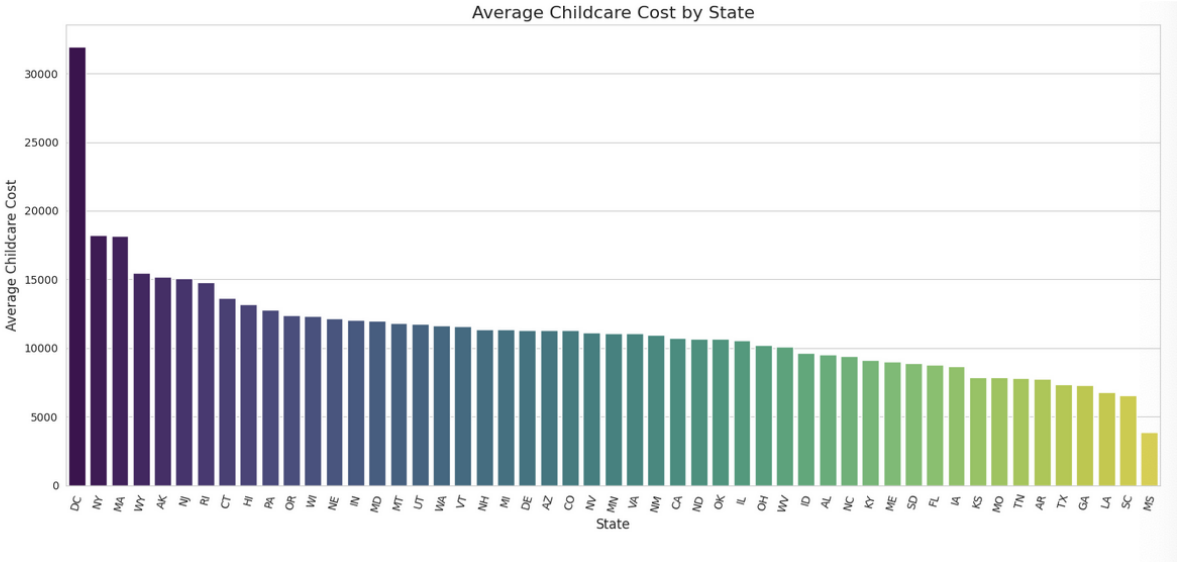
This chart shows the average cost of transportation from state to state. The states are organized from most expensive (Utah) to least expensive (Washington DC). Allows for easy comparison between each state and how the cost trends from state to state.

Average Healthcare Cost by State:



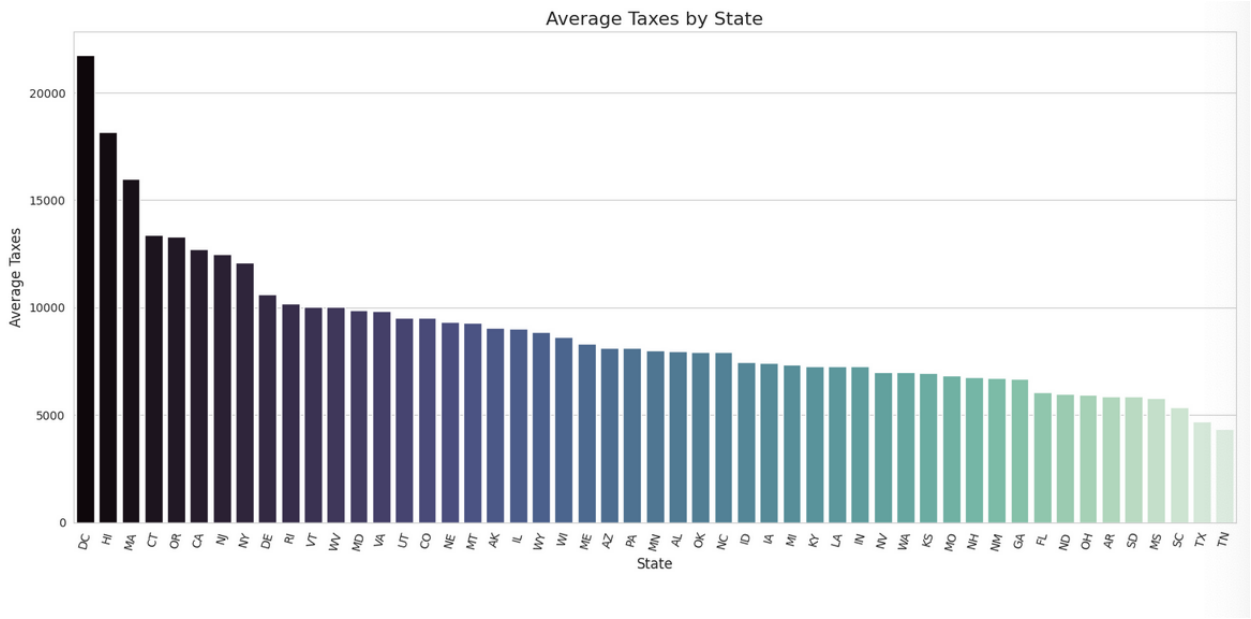
Similar to the transportation graph, this shows the trends from state to state for healthcare cost. This graph is not as evenly distributed. West Virginia has a substantially higher cost of healthcare than all other states

Average Childcare Cost by State:



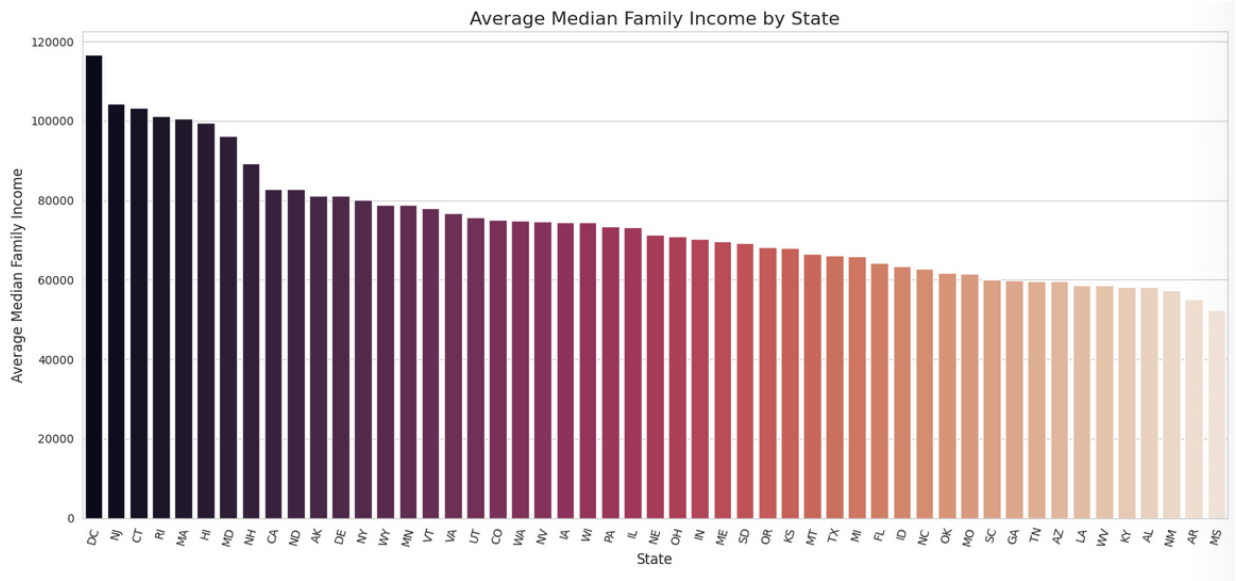
This chart has a noticeable outlier, Washington DC, with the childcare cost being almost double the cost of the next highest state. The states with the most expensive childcare costs are typically states known for being expensive to live in, typically in the North East region of the country. On the other end, the states with cheaper childcare are mostly southern states known for a lowk cost of living.

Average Taxes by State:



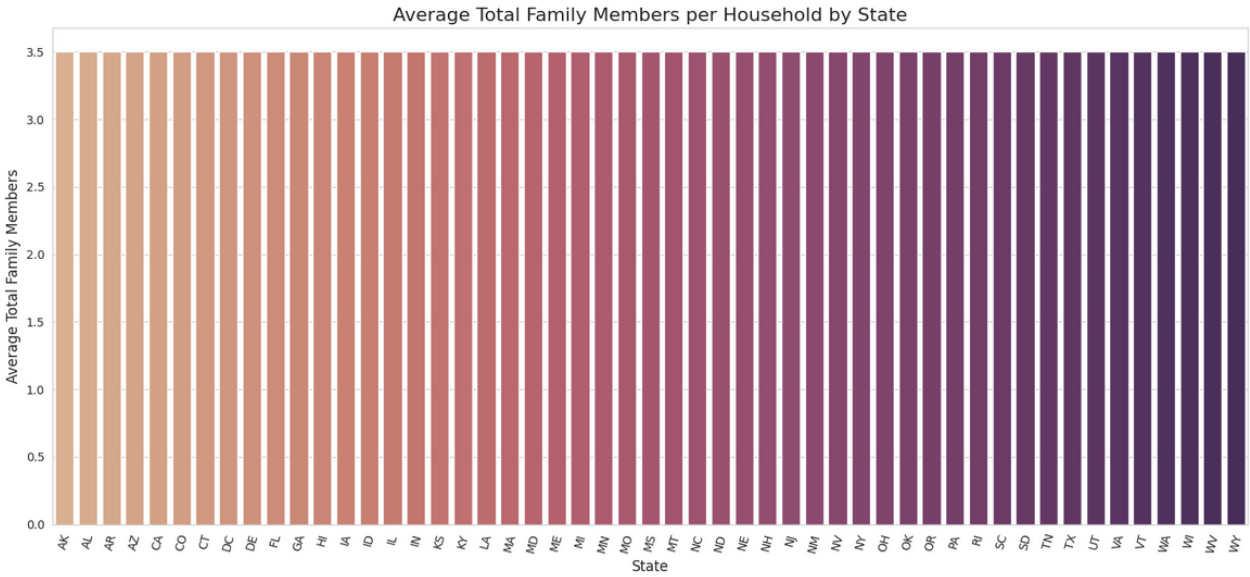
This follows a similar pattern to the childcare cost by state graph. Washington DC is the most taxed state by a large margin. The other higher-taxed states are once again in the Northeast. On the other side, the lower-taxed states are mostly in the South.

Average Median Family Income by State:



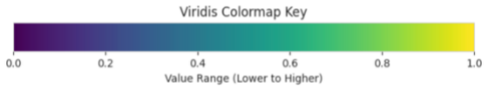
Once again Washington DC leads the pack with the most income by state. This follows the same pattern as the previous two graphs. The higher-income states are in the Northeast and the lower-income states are in the South. This is leads one to believe there is a correlation between income, costs, and state.

Average Total Family Members per Household by State:



*****Something wrong with this chart, I need to fix it. *****

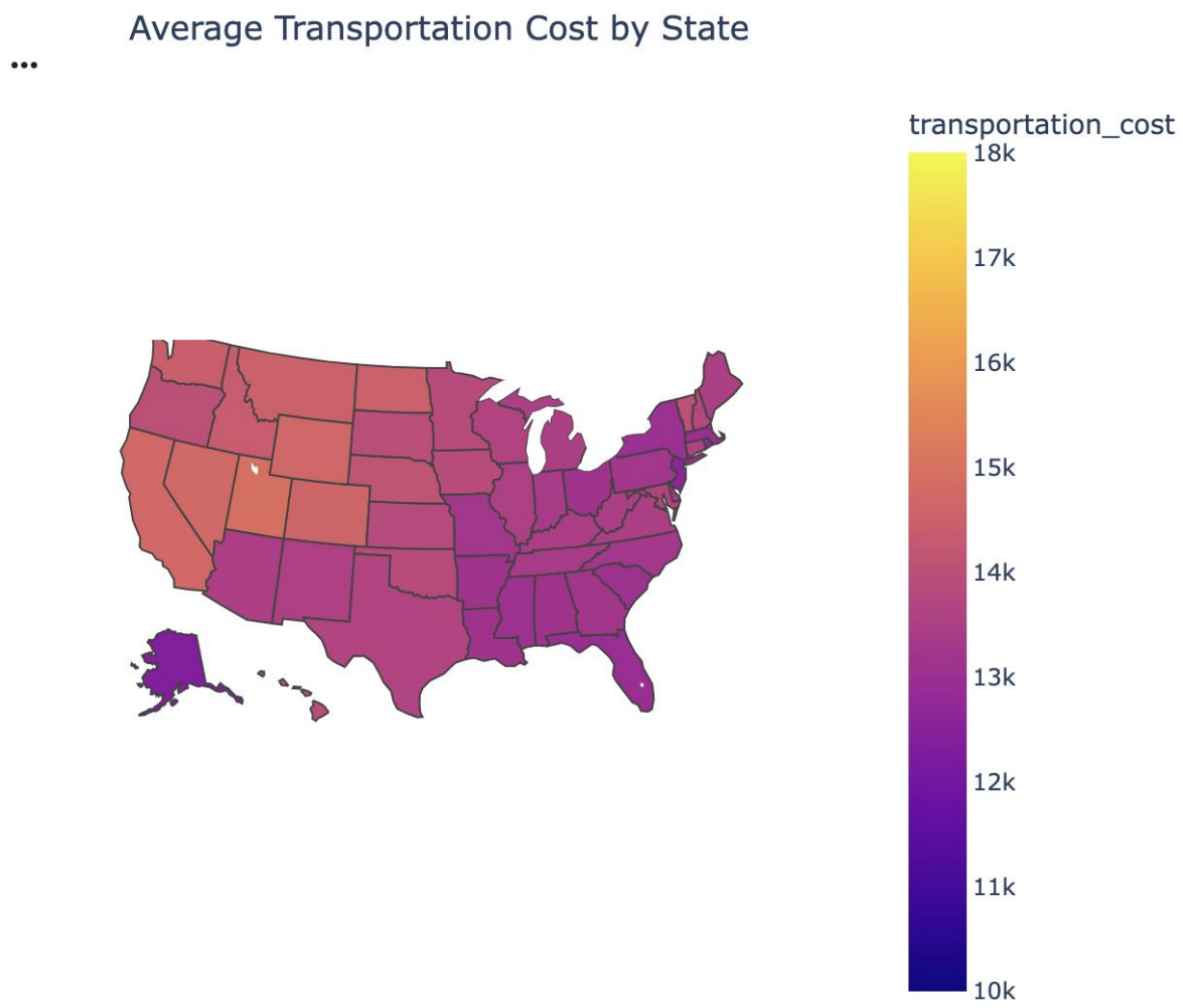
	state	total_cost	median_family_income
0	AK	85234.557360	81242.831358
1	AL	68967.570020	58099.871502
2	AR	61491.891851	55025.567812
3	AZ	75985.939376	59661.879167
4	CA	93179.163060	82922.169248
5	CO	77595.782964	75158.805298
6	CT	90013.504620	103228.233398
7	DC	120289.636920	116607.390625
8	DE	81505.517560	81198.664062
9	FL	71521.722981	64294.176597
10	GA	64898.336594	59841.336134
11	HI	104897.262264	99596.992188
12	IA	66089.039308	74522.305003
13	ID	70506.019440	63504.196911
14	IL	73024.613053	73194.425781
15	IN	69161.680460	70347.774541
16	KS	67233.076625	68094.712537
17	KY	65294.227775	58144.298665
18	LA	68282.349846	58620.873138
19	MA	101891.575946	100532.209821
20	MD	79001.490465	96138.127604
21	ME	74208.436575	69733.939209
22	MI	68782.098743	66016.922863
23	MN	71748.349877	78767.613775
24	MO	66123.685410	61569.829042
25	MS	61360.990604	52281.524319
26	MT	74756.847354	66608.078753
27	NC	70013.586953	62694.322305
28	ND	68709.927609	82881.291421
29	NE	77684.601942	71279.636341
30	NH	74802.494172	89199.250000
31	NJ	92575.263891	104246.459821
32	NM	67687.562087	57366.033144
33	NV	75925.310202	74772.627068
34	NY	86764.054353	80175.488344
35	OH	65100.187508	70903.528720
36	OK	71110.675373	61805.006494
37	OR	83818.032683	68232.733398
38	PA	72114.340764	73501.250641
39	RI	83512.769736	101153.248438
40	SC	62104.869806	60021.241338
41	SD	70538.723187	69327.778646
42	TN	62871.107261	59715.511924
43	TX	64795.398352	66050.294268
44	UT	75609.054360	75671.131061
45	VA	76748.558009	76830.189527
46	VT	83138.324486	77969.394531
47	WA	75648.910151	74820.338141
48	WI	73150.619482	74495.815864
49	WV	76899.703265	58596.519176
50	WY	84289.773668	78918.055027



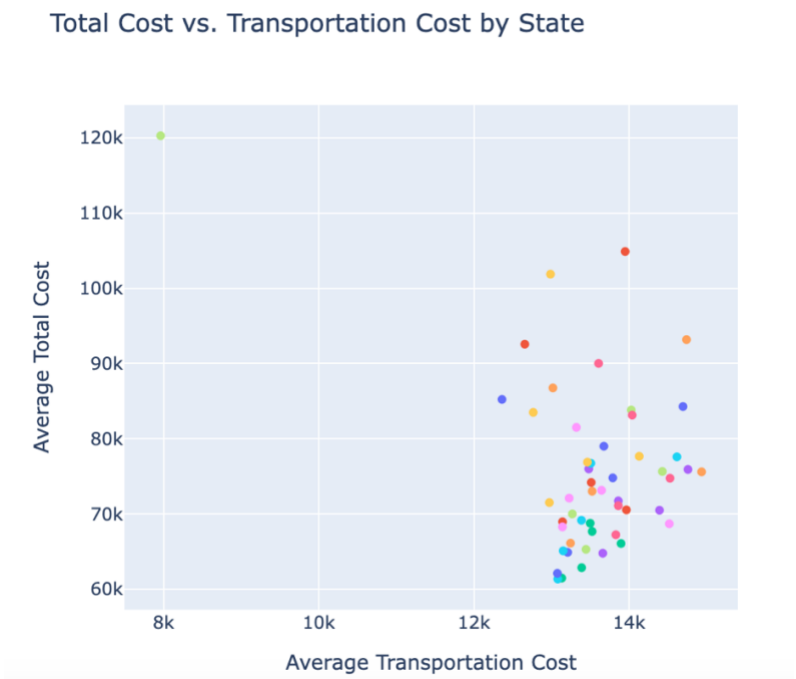
Total_cost and Median_Family_Income by State

The shading allows for the extreme values on both sides to be easily identified. As seen in the previous graphs, the chart confirms that DC is the most expensive state but also has the highest income per family. On the contrary, states like Mississippi and Texas have low costs but also low income. This table also makes it easy to notice the correlation between income and expenses as states with high-income families also tend to have higher costs.

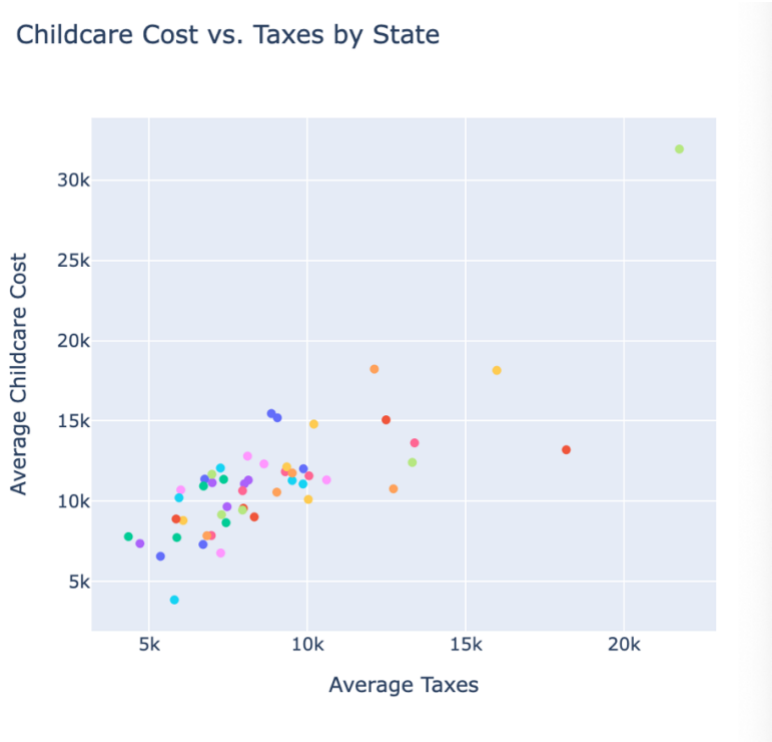
Average Transportation Cost by State (Choropleth):



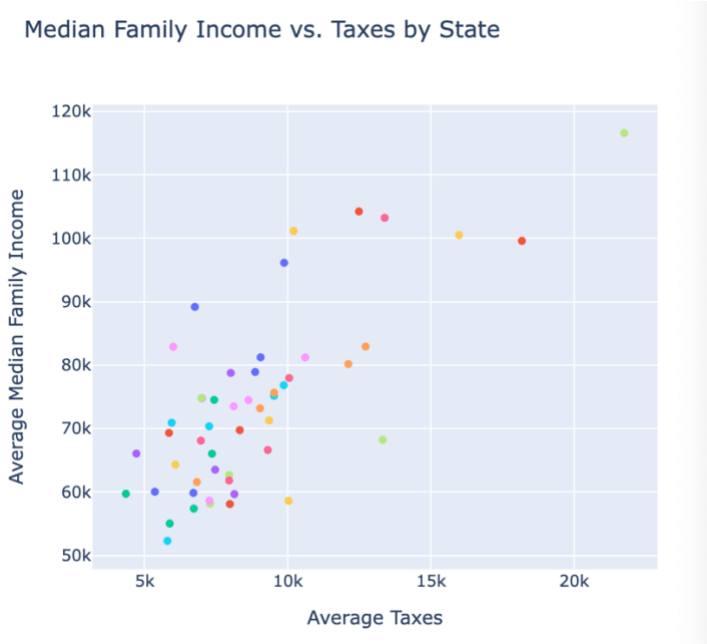
This map offers a nationwide view on how transportation cost varies by state. This provides a simpler visual than a chart or graph because it is easier to see trends in transportation cost across the state. Moving more towards the east, transportation cost get less expensive. This can be seen because the color gradient gets darker, showing a decrease in cost. This is likely because transportation such as subways, trains, and taxis are more prevalent and accessible in the northeast cities due to their higher densities of people. Matches the results in #3, but this provides a better visual on how the cost varies region to region. However, this map makes it hard to see smaller states such as Washington Dc and Rhode Island.



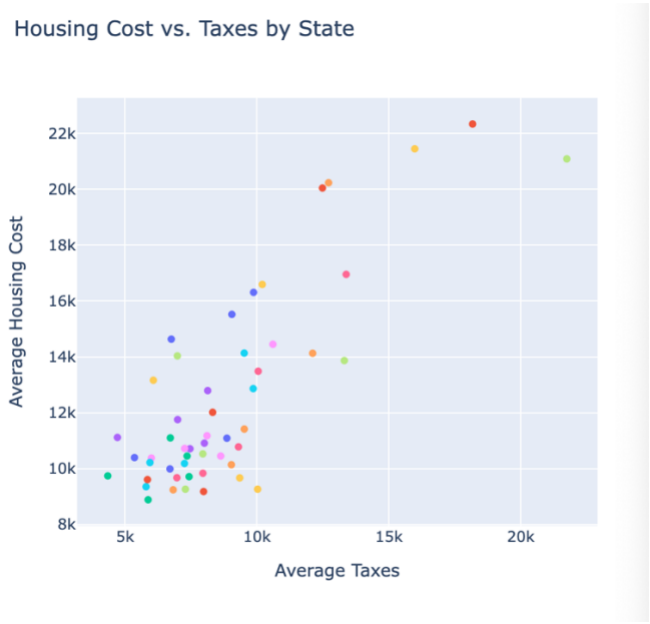
This shows a slight positive correlation between Average Total Cost and Average Transportation Cost, meaning that states that have a higher cost in one will likely have a higher cost in the other. The point in the top left of the graph is Washington DC which stands out as an outlier.



This plot shows a very apparent positive correlation between average taxes and average childcare cost. This means states with higher taxes have a higher childcare cost and vice versa.



This plot also shows a positive correlation between Average Taxes and Average Median Family Income. Families that have a higher income pay more in taxes. Once again, Washington DC is an outlier which can be seen in the top right corner. Meaning, DC has the highest taxes and highest average median income.



This plot shows another positive correlation. This one is between Average Housing Cost and Average Taxes. Logically, this correlation makes sense because if a house is more expensive, then the taxes will be higher.

12 Most Expensive Counties for Childcare

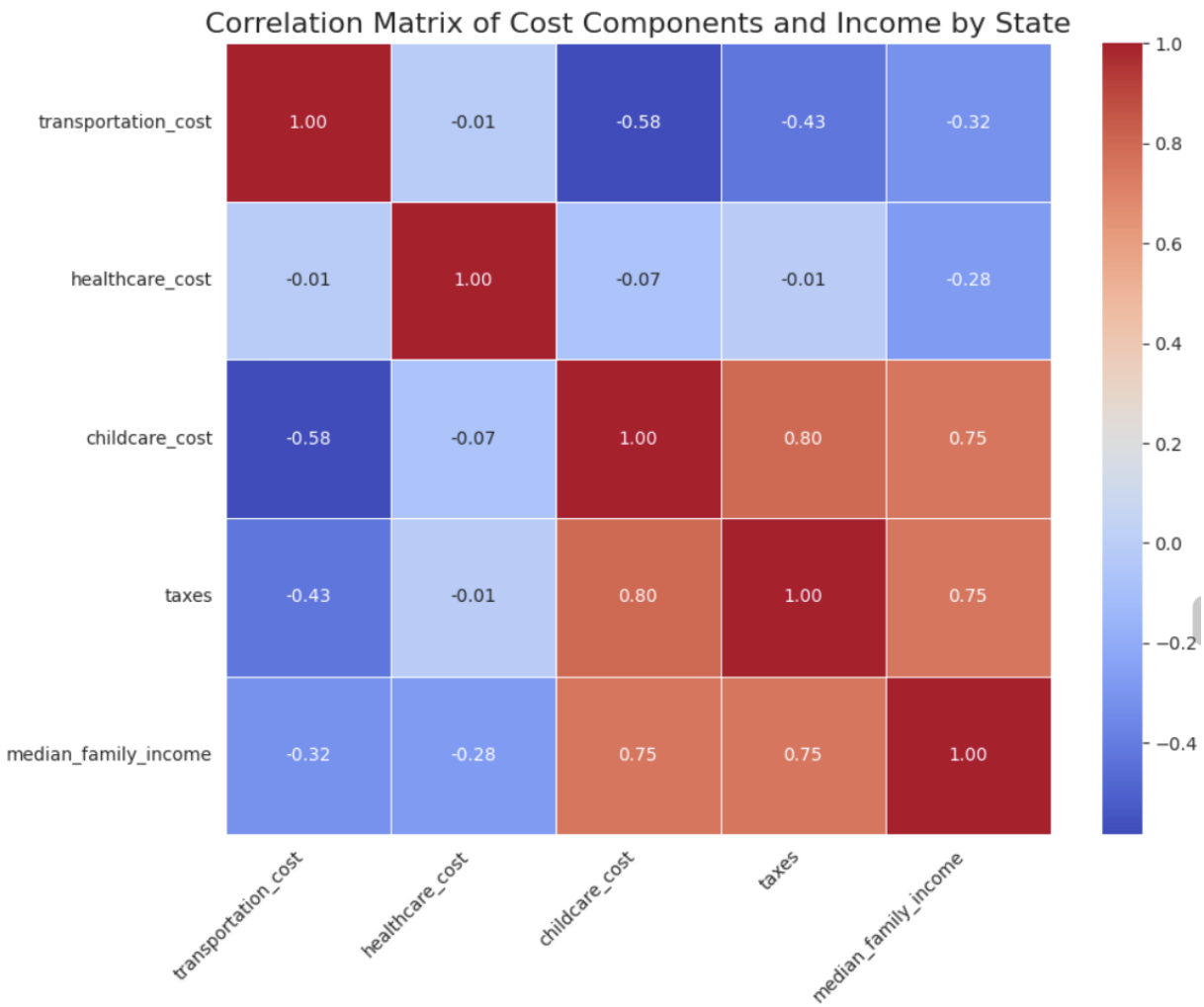
	county	childcare_cost
493	District of Columbia	31940.492400
1187	New York County	29620.246800
1379	Queens County	28050.334320
1439	Rockland County	27260.377200
1796	Westchester County	26661.030720
1617	Suffolk County	25288.633680
206	Bronx County	24340.542960
1203	Norfolk County	23606.173680
519	Dutchess County	22844.859600
1714	Ulster County	21947.080944
1677	Tompkins County	21589.945152
670	Grand Forks County	21525.153984

12 Most Expensive Counties for Taxes

	county	taxes
1480	San Mateo County	34738.964880
1043	Marin County	31637.529360
1475	San Francisco County	27174.135960
1187	New York County	26517.054360
1492	Santa Clara County	24651.857760
1439	Rockland County	23463.096600
1379	Queens County	22807.598208
1203	Norfolk County	22347.509448
1796	Westchester County	22292.750676
1617	Suffolk County	22256.451666
493	District of Columbia	21744.785220
779	Honolulu County	21423.277428

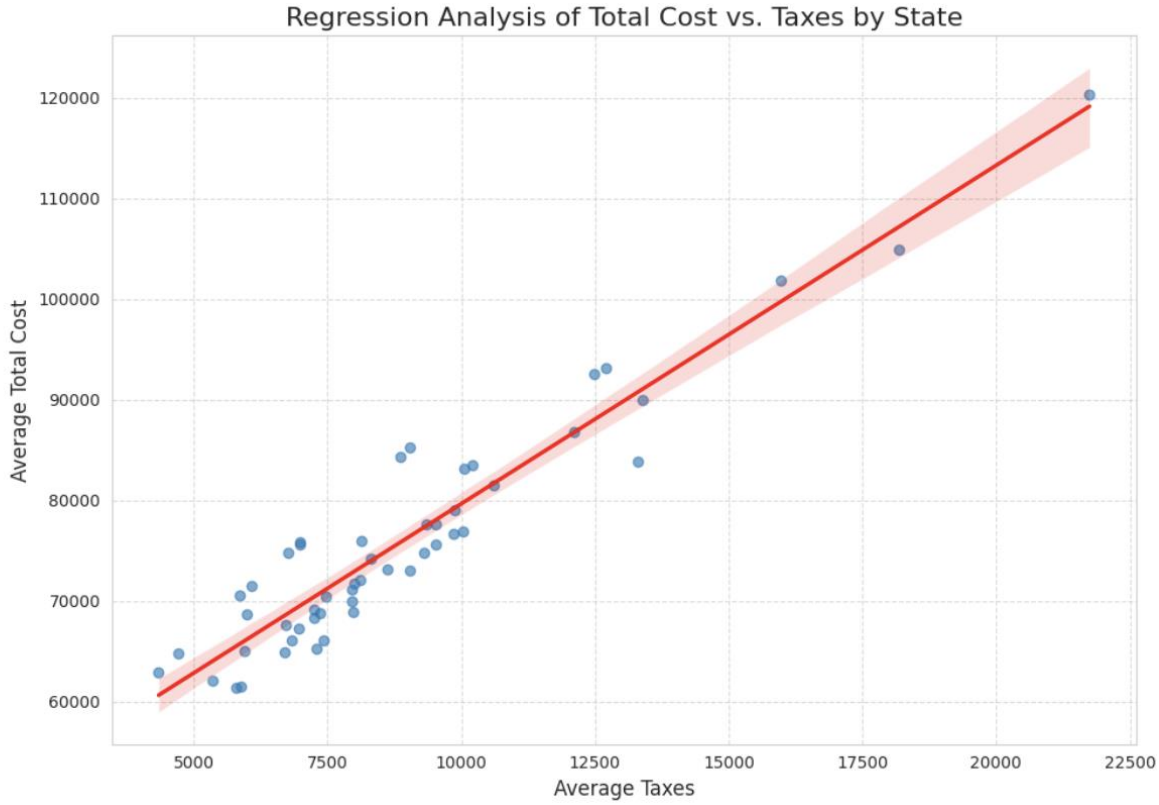
Several counties appear on both charts. Ex: District of Columbia, Norfolk, New York County, etc. These counties also come from similar regions, the Northeast. This lines up with the trends and patterns seen in the previous graphs and maps.

Correlation Matrix:



This correlation matrix confirms what has been interpreted through the previous graphs and charts. The variables that are most noticeably correlated with one another are the income and expenses variables. Specifically, the bottom right corner of the correlation matrix is almost all red, meaning there is a correlation.

Total Cost vs Taxes by State Regression Analysis:

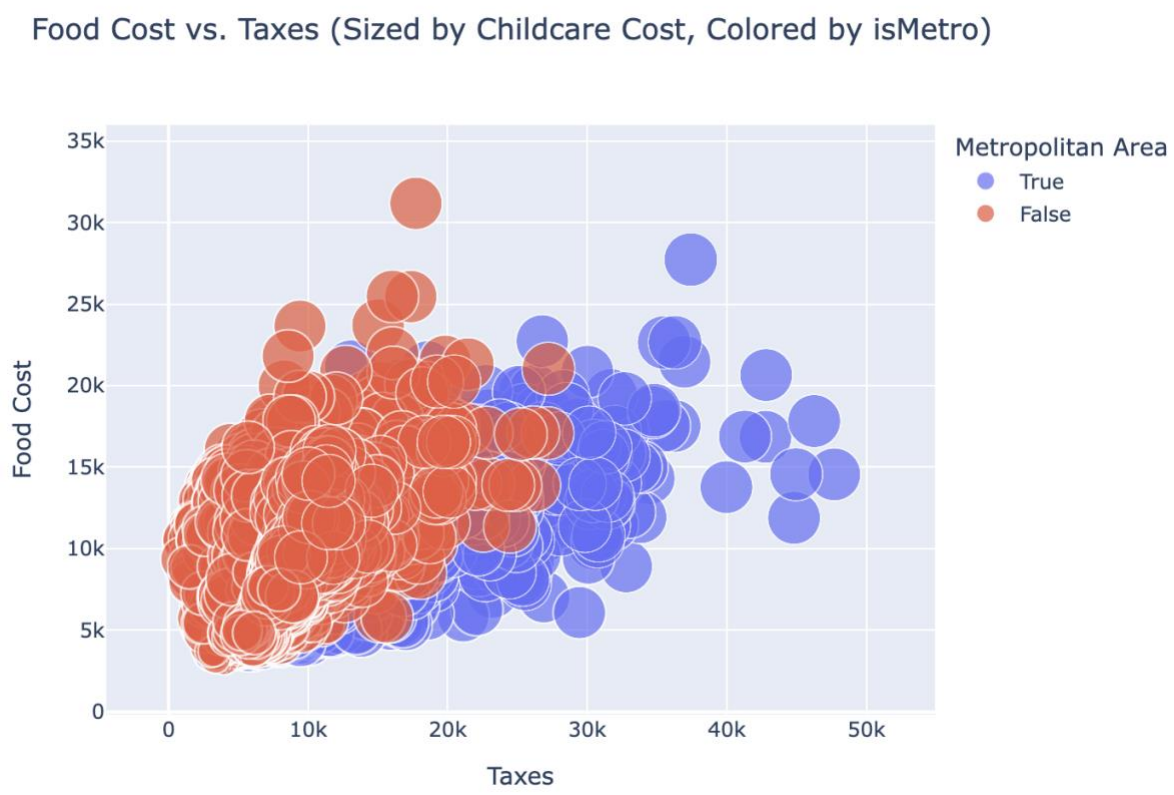


This graph offers another visual showing the correlation between Average Taxes and Average Total Cost. Majority of data points fit close to the regression line showing a positive correlation between taxes and total cost. Even the outliers in the top right of the graph still are on the regression line.

Outliers for food_cost by State:

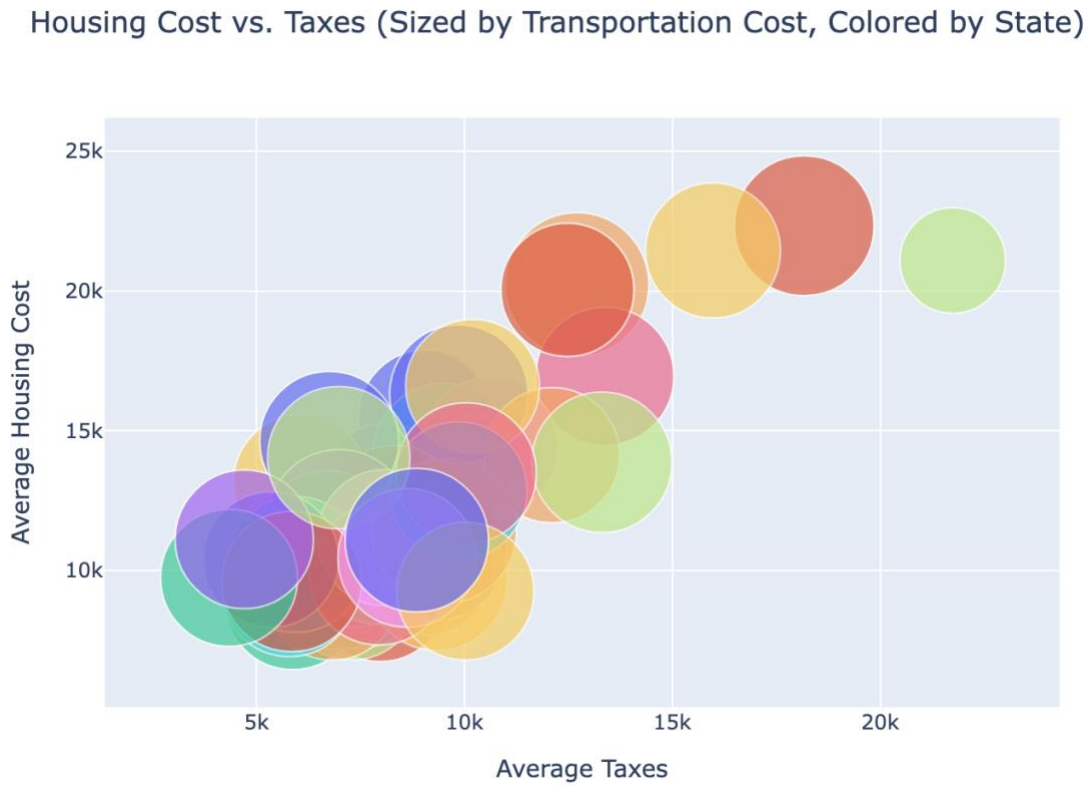
After using the IQR method to find outliers for food cost by state, 66 outliers were found in the data. The lower bound for outliers was -1551.88. Any value below this number was considered an outlier. The up[per bound was 18056.92. Any value above this number was considered an outlier.

Food Cost vs. Taxes (Bubble Plot):

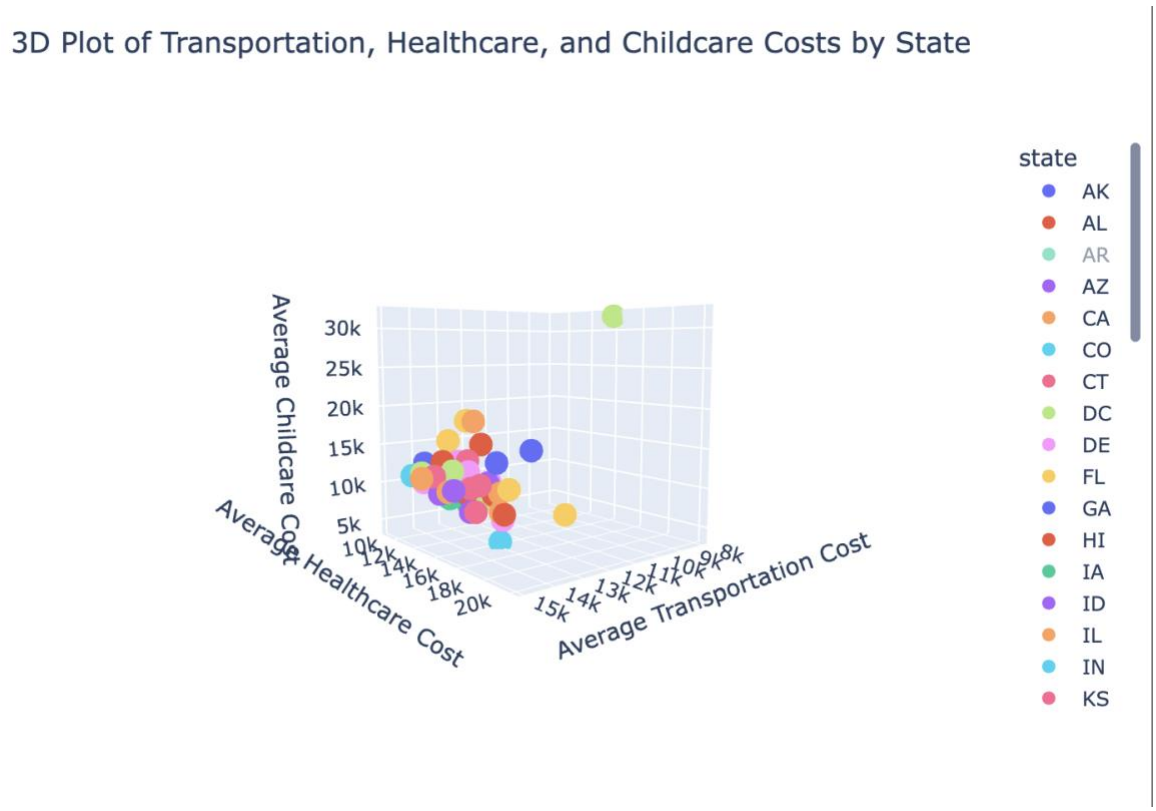


This bubble plot shows a slight positive correlation between taxes and food cost. However, there is a noticeable correlation between food cost, taxes, and whether or not the state is metro or not. States that are metro tend to have higher taxes and food costs.

Housing Costs vs. Taxes (Bubble Plot)



This bubble plot shows a positive correlation between average housing cost and average taxes. However, this correlation is mostly due to the more extremes of each end as there is a large clump of states at the bottom.



This graph offers a 3D visual of Childcare Cost, Healthcare Cost, and transportation Cost. Through this visual, there is an apparent outlier, being Washington DC again. There is not much correlation between any of the variables; most of the data points are clumped together with no relationship being shared.

3D Plot of Childcare Cost. Taxes and Median Family Income by State:

Without interacting with the graph, this graph is hard to interpret, but when you look at it, you can see a positive correlation between average median income and average taxes. The same correlation can be seen for median family income and average childcare cost.

Principal Component Analysis (PCA):

	principal_component_1	principal_component_2	principal_component_3	state
0	1.430916	1.472689	-0.264299	AK
1	-1.399681	0.783368	-0.310554	AL
2	-2.400937	-0.087592	-1.125436	AR
3	-0.701385	0.752184	0.098576	AZ
4	2.233578	-0.577477	2.143729	CA

```
Explained variance ratio by each component: [0.6478631  0.14671901 0.10645242]
Total explained variance: 0.90
```

Non-negative Matrix Factorization (NMF):

	nmf_component_1	nmf_component_2	nmf_component_3	state
0	0.692317	0.469774	0.428719	AK
1	0.069112	0.398688	0.506586	AL
2	0.000000	0.213107	0.506744	AR
3	0.262135	0.413990	0.527317	AZ
4	1.039016	0.336281	0.654635	CA

```
NMF components (transformed data) for the first 5 states:
```

Conclusion

Through various data visualizations, it was clear that several correlations can be found in this data set. The most obvious ones were the correlation of several different income-related variables and the various expense variables. The data showed that areas with a higher median income also had higher expenses. These expenses varied from transportation to housing to food to taxes. Washington DC stood out as the place that tended to have the highest income and expenses.

Another pattern that stood out in the graphs and was then confirmed in the choropleth map of the US was that the farther east states tended to be the most expensive to live. Most big cities in the US are on the East Coast, which is why transportation and taxes were typically higher in this region.

Other variables that showed some kind of correlation were taxes and whether or not the city was metro, housing cost and taxes, median family income, and childcare cost, just to name a few. Median income and childcare cost stood out because at first, it would not seem like they would have much in common. But, families that have higher incomes are able to afford the burden of paying for the costs that come along with having a child, which means they are more likely to have children.

While this dataset comes with several variables, most of these variables can be narrowed down to income and expenses. Looking beyond that basic comparison, geographical patterns begin to appear in the data that help interpret the data through a different lens. The basic conclusions from this data is that families that have higher income tend to have more expenses as well, and states on the East Coast tend to be more expensive to live in than states in the West and the South.

Appendix

```
import pandas as pd

1) Clean the dataset

df = pd.read_csv('/content/cost_of_living_us.csv')
display(df.head())

print(f"Shape of DataFrame before dropping nulls: {df.shape}")
df.dropna(inplace=True)
print(f"Shape of DataFrame after dropping nulls: {df.shape}")
display(df.head())

summarized_df = df.groupby('state')[['total_cost', 'median_family_income']].mean().reset_index()
display(summarized_df)

### Analysis of Costs, Income, and Family Size by State
import matplotlib.pyplot as plt
import seaborn as sns
import re

# --- Prepare Data for Numerical Columns ---

numerical_columns = [
    'transportation_cost',
    'healthcare_cost',
    'childcare_cost',
    'taxes',
    'median_family_income'
]

# Aggregate data: mean of specified numerical columns by state
```

```
state_avg_costs_income = df.groupby('state')[numerical_columns].mean().reset_index()

# --- Prepare Data for family_member_count ---

def parse_family_member_count(s):
    """Parses a string like '1p0c' to return the total number of family members."""
    match = re.match(r'(\d+)p(\d+)c', str(s))
    if match:
        parents = int(match.group(1))
        children = int(match.group(2))
        return parents + children
    return None # Handle cases that don't match the pattern

df['total_family_members'] = df['family_member_count'].apply(parse_family_member_count)

# Aggregate average total family members by state
state_avg_family_members = df.groupby('state')['total_family_members'].mean().reset_index()

# Set up matplotlib plotting style
sns.set_style("whitegrid")

#### Average Transportation Cost by State
plt.figure(figsize=(15, 7))
sns.barplot(x='state', y='transportation_cost',
data=state_avg_costs_income.sort_values(by='transportation_cost', ascending=False), palette='magma')
plt.title('Average Transportation Cost by State', fontsize=16)
plt.xlabel('State', fontsize=12)
plt.ylabel('Average Transportation Cost', fontsize=12)
plt.xticks(rotation=75)
plt.tight_layout()
plt.show()

#### 2) Average Healthcare Cost by State
```

```
plt.figure(figsize=(15, 7))

sns.barplot(x='state', y='healthcare_cost', data=state_avg_costs_income.sort_values(by='healthcare_cost', ascending=False), palette='crest')

plt.title('Average Healthcare Cost by State', fontsize=16)

plt.xlabel('State', fontsize=12)

plt.ylabel('Average Healthcare Cost', fontsize=12)

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()


plt.figure(figsize=(15, 7))

sns.barplot(x='state', y='childcare_cost', data=state_avg_costs_income.sort_values(by='childcare_cost', ascending=False), palette='viridis')

plt.title('Average Childcare Cost by State', fontsize=16)

plt.xlabel('State', fontsize=12)

plt.ylabel('Average Childcare Cost', fontsize=12)

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()



plt.figure(figsize=(15, 7))

sns.barplot(x='state', y='taxes', data=state_avg_costs_income.sort_values(by='taxes', ascending=False), palette='mako')

plt.title('Average Taxes by State', fontsize=16)

plt.xlabel('State', fontsize=12)

plt.ylabel('Average Taxes', fontsize=12)

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()


#### Average Median Family Income by State
```

```
plt.figure(figsize=(15, 7))

sns.barplot(x='state', y='median_family_income',
data=state_avg_costs_income.sort_values(by='median_family_income', ascending=False), palette='rocket')

plt.title('Average Median Family Income by State', fontsize=16)

plt.xlabel('State', fontsize=12)

plt.ylabel('Average Median Family Income', fontsize=12)

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()


#### Average Total Family Members per Household by State

plt.figure(figsize=(15, 7))

sns.barplot(x='state', y='total_family_members',
data=state_avg_family_members.sort_values(by='total_family_members', ascending=False), palette='flare')

plt.title('Average Total Family Members per Household by State', fontsize=16)

plt.xlabel('State', fontsize=12)

plt.ylabel('Average Total Family Members', fontsize=12)

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()


#### 3) Summarized Data by State with Color-Coded Values


styled_summarized_df = summarized_df.style.background_gradient(subset=['total_cost',
'median_family_income'], cmap='viridis')

display(styled_summarized_df)


#### Color Key for Table Gradients


import numpy as np

import matplotlib.colors as mcolors


# Create a figure and an axis
```

```
fig, ax = plt.subplots(figsize=(8, 1))
fig.subplots_adjust(bottom=0.5)

# Create a colormap instance
cmap = plt.get_cmap('viridis')
norm = mcolors.Normalize(vmin=0, vmax=1)

# Create a colorbar
cbar = fig.colorbar(plt.cm.ScalarMappable(cmap=cmap, norm=norm),
                    cax=ax, orientation='horizontal')

cbar.set_label('Value Range (Lower to Higher)')
plt.title('Viridis Colormap Key')
plt.show()

#### 4) Analyzing Relationships Between Cost Components and Income by State

import plotly.express as px

# Calculate average housing_cost by state
state_avg_housing_cost = df.groupby('state')['housing_cost'].mean().reset_index()

# Merge summarized_df (for total_cost) into state_avg_costs_income
combined_state_data = pd.merge(state_avg_costs_income, summarized_df[['state', 'total_cost']], on='state',
how='left')

# Merge state_avg_housing_cost into combined_state_data
combined_state_data = pd.merge(combined_state_data, state_avg_housing_cost, on='state', how='left')

display(combined_state_data.head())

##### 5) Total Cost vs. Transportation Cost by State
```

```
import plotly.express as px

fig = px.choropleth(
    state_avg_costs_income,
    locations='state',
    locationmode='USA-states',
    color='transportation_cost',
    scope='usa',
    color_continuous_scale='Plasma',
    range_color=[10000, 18000],  # Adjusted range for a smaller key
    title='Average Transportation Cost by State',
    hover_name='state',
    hover_data={'transportation_cost': True}
)
fig.update_layout(
    geo_scope='usa',
    margin={"r":0,"t":50,"l":0,"b":0}
)
fig.show()
```

6) Plots for:

a. total_cost vs. transportation_cost

b. childcare_cost vs. taxes

c. median_family_income vs. taxes

d. housing cost vs. taxes

```
fig_a = px.scatter(combined_state_data,
                   x='transportation_cost',
                   y='total_cost',
                   color='state',
                   hover_name='state',
```

```
            title='Total Cost vs. Transportation Cost by State',
            labels={'transportation_cost': 'Average Transportation Cost', 'total_cost': 'Average Total Cost'})
fig_a.update_layout(showlegend=False) # Hide legend as hover provides state info
fig_a.show()


##### b. Childcare Cost vs. Taxes by State

fig_b = px.scatter(combined_state_data,
            x='taxes',
            y='childcare_cost',
            color='state',
            hover_name='state',
            title='Childcare Cost vs. Taxes by State',
            labels={'taxes': 'Average Taxes', 'childcare_cost': 'Average Childcare Cost'})
fig_b.update_layout(showlegend=False)
fig_b.show()


##### c. Median Family Income vs. Taxes by State

fig_c = px.scatter(combined_state_data,
            x='taxes',
            y='median_family_income',
            color='state',
            hover_name='state',
            title='Median Family Income vs. Taxes by State',
            labels={'taxes': 'Average Taxes', 'median_family_income': 'Average Median Family Income'})
fig_c.update_layout(showlegend=False)
fig_c.show()
```

```
##### d. Housing Cost vs. Taxes by State

fig_d = px.scatter(combined_state_data,
                   x='taxes',
                   y='housing_cost',
                   color='state',
                   hover_name='state',
                   title='Housing Cost vs. Taxes by State',
                   labels={'taxes': 'Average Taxes', 'housing_cost': 'Average Housing Cost'})
fig_d.update_layout(showlegend=False)
fig_d.show()

#### 7) Top 12 Most Expensive Counties for Childcare Cost


county_avg_childcare_cost = df.groupby('county')['childcare_cost'].mean().reset_index()
top_12_childcare_counties = county_avg_childcare_cost.sort_values(by='childcare_cost',
ascending=False).head(12)
display(top_12_childcare_counties)

#### 8) Top 12 Most Expensive Counties for Taxes

county_avg_taxes = df.groupby('county')['taxes'].mean().reset_index()
top_12_tax_counties = county_avg_taxes.sort_values(by='taxes', ascending=False).head(12)
display(top_12_tax_counties)

#### 9) Correlation Matrix of Cost Components and Income by State

import seaborn as sns
import matplotlib.pyplot as plt

# Calculate the correlation matrix for the numerical columns in state_avg_costs_income
correlation_matrix = state_avg_costs_income[numerical_columns].corr()
```

```
# Set up the matplotlib figure
plt.figure(figsize=(10, 8))

# Draw the heatmap with the mask and correct aspect ratio
sns.heatmap(correlation_matrix,
            annot=True, # Show the correlation values on the heatmap
            cmap='coolwarm', # Choose a diverging colormap
            fmt=".2f", # Format annotation to two decimal places
            linewidths=.5) # Add lines between cells

plt.title('Correlation Matrix of Cost Components and Income by State', fontsize=16)
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

####
10) Regression Analysis: Total Cost vs. Taxes by State



plt.figure(figsize=(10, 7))
sns.regplot(x='taxes', y='total_cost', data=combined_state_data, scatter_kws={'alpha':0.6},
line_kws={'color':'red'})
plt.title('Regression Analysis of Total Cost vs. Taxes by State', fontsize=16)
plt.xlabel('Average Taxes', fontsize=12)
plt.ylabel('Average Total Cost', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()



state_avg_food_cost = df.groupby('state')['food_cost'].mean().reset_index()
```

```
combined_state_data = pd.merge(combined_state_data, state_avg_food_cost, on='state', how='left')
display(combined_state_data.head())
```

11) Outlier Detection for Food Cost using IQR Method

```
# Calculate Q1 (25th percentile) and Q3 (75th percentile) for 'food_cost'
Q1 = df['food_cost'].quantile(0.25)
Q3 = df['food_cost'].quantile(0.75)

# Calculate the Interquartile Range (IQR)
IQR = Q3 - Q1

# Define the outlier fences
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print(f"Q1 (25th percentile): {Q1:.2f}")
print(f"Q3 (75th percentile): {Q3:.2f}")
print(f"IQR: {IQR:.2f}")
print(f"Lower Bound for Outliers: {lower_bound:.2f}")
print(f"Upper Bound for Outliers: {upper_bound:.2f}")

# Identify outliers
outliers = df[(df['food_cost'] < lower_bound) | (df['food_cost'] > upper_bound)]

display(f"Number of outliers found: {len(outliers)}")
display("First 10 rows of food_cost outliers:")
display(outliers.head(10))
```

12) Bubble Plot: Food Cost vs. Taxes, Sized by Childcare Cost and Colored by Metropolitan Status

```
fig = px.scatter(df, # Use the original df to include 'isMetro' for individual points
```

```
        x='taxes',
        y='food_cost',
        size='childcare_cost',
        color='isMetro',
        hover_name='county',
        title='Food Cost vs. Taxes (Sized by Childcare Cost, Colored by isMetro)',
        labels={'taxes': 'Taxes', 'food_cost': 'Food Cost', 'childcare_cost': 'Childcare Cost', 'isMetro':
'Metropolitan Area'},
        log_x=False, # Set to True if tax distribution is highly skewed
        size_max=60) # Adjust maximum bubble size for better visualization

fig.update_layout(showlegend=True)
fig.show()
```

13) Bubble Chart for Housing vs. Taxes

```
fig = px.scatter(combined_state_data,
        x='taxes',
        y='housing_cost',
        size='transportation_cost',
        color='state',
        hover_name='state',
        title='Housing Cost vs. Taxes (Sized by Transportation Cost, Colored by State)',
        labels={'taxes': 'Average Taxes', 'housing_cost': 'Average Housing Cost', 'transportation_cost':
'Average Transportation Cost'},
        size_max=60) # Adjust maximum bubble size for better visualization

fig.update_layout(showlegend=False) # Hide legend as hover provides state info
fig.show()
```

14) 3D Plot for transportation, healthcare_cost, and childcare_cost

```
fig = px.scatter_3d(combined_state_data,
                x='transportation_cost',
                y='healthcare_cost',
                z='childcare_cost',
                color='state',
                hover_name='state',
                title='3D Plot of Transportation, Healthcare, and Childcare Costs by State',
                labels={'transportation_cost': 'Average Transportation Cost',
                        'healthcare_cost': 'Average Healthcare Cost',
                        'childcare_cost': 'Average Childcare Cost'})
fig.show()
```

15) 3D Plot for childcare_cost, taxes, median_family_income, and state

```
fig = px.scatter_3d(combined_state_data,
                x='childcare_cost',
                y='taxes',
                z='median_family_income',
                color='state',
                hover_name='state',
                title='3D Plot of Childcare Cost, Taxes, and Median Family Income by State',
                labels={'childcare_cost': 'Average Childcare Cost',
                        'taxes': 'Average Taxes',
                        'median_family_income': 'Average Median Family Income'})
fig.show()
```

16) PCA for 3 components

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA

# Select the numerical columns for PCA
pca_features = [
    'transportation_cost',
    'healthcare_cost',
    'childcare_cost',
    'taxes',
    'median_family_income',
    'total_cost',
    'housing_cost',
    'food_cost'
]

x = combined_state_data[pca_features].values

# Standardize the features
x_scaled = StandardScaler().fit_transform(x)

# Apply PCA with 3 components
pca = PCA(n_components=3)
principal_components = pca.fit_transform(x_scaled)

# Create a DataFrame with the principal components
pca_df = pd.DataFrame(data = principal_components,
                      columns = ['principal_component_1', 'principal_component_2', 'principal_component_3'])

# Add the state names back for identification
pca_df['state'] = combined_state_data['state']
display(pca_df.head())

print(f"Explained variance ratio by each component: {pca.explained_variance_ratio_}")
```

```
print(f"Total explained variance: {pca.explained_variance_ratio_.sum():.2f}")
```

17) NMF for Three Components

```
#### Non-negative Matrix Factorization (NMF) for 3 Components

from sklearn.decomposition import NMF
from sklearn.preprocessing import MinMaxScaler

x_nmf_scaled = MinMaxScaler().fit_transform(x)

nmf = NMF(n_components=3, init='random', random_state=42, max_iter=1000)
nmf_components = nmf.fit_transform(x_nmf_scaled)

nmf_df = pd.DataFrame(data = nmf_components,
                      columns = ['nmf_component_1', 'nmf_component_2', 'nmf_component_3'])

nmf_df['state'] = combined_state_data['state']
display(nmf_df.head())
print("NMF components (transformed data) for the first 5 states:")
```